PAYMENTS INNOVATION TRIBE

AI Use Cases
Payments & Treasury

Multi-Agent Orchestration for SWIFT Payment Processing & Fraud Detection

Interest Areas for Investments

	Domestic Interbank	Cross Border Interbank
Treasury	Intraday Liquidity & Collateral Optimization Cash forecasting & working capital co-pilot	FX Optimization & Execution
Operations	Agentic routing & rail selections (spans both) Autonomous exception handling and repair (spans both) Nostro,/RTGS account reconciliation	Intelligent Claims & Investigations Interbank fee, pricing, and SLA optimization

DEEP DIVE

AI-Native
Claims & Investigation System

Multi-Agent Orchestration for SWIFT Payment Processing & Fraud Detection

The Challenge

Why Traditional Claims Processing Falls Short

Manual Processing Bottlenecks

Average claim resolution takes 5-7 days with 60% requiring human intervention

Siloed Decision Making

Fraud, compliance, and validation teams work in isolation, missing cross-domain patterns

Evolving Fraud Patterns

Static rules cannot adapt to sophisticated, ever-changing fraud schemes

$32B

Global payment fraud losses annually

15%

False positive rate in legacy systems

72hrs

Average time to detect sophisticated fraud

"Financial institutions need intelligent systems that learn, adapt, and collaborate in real-time."

The Solution

AI-Native Multi-Agent Architecture

8 Specialized AI Agents with Vector Search and Graph Analysis

Intelligent Detection

- Vector embeddings for patterns
- Graph analysis for fraud
- Real-time anomaly detection
- Continuous learning

Compliance First

- ISO 20022 / SWIFT compliant
- AML/Sanctions/PEP screening
- Complete audit trails
- Human-in-the-loop controls

Enterprise Ready

- Sub-30 second processing
- 99.9% system availability
- Real-time WebSocket updates
- Comprehensive API layer

95%+	5%-	30s	8
Detection Rate	False Positives	Processing	AI Agents

System Architecture

Layered Design for Scalability and Performance

PRESENTATION

Enterprise Dashboard

Real-time WebSocket updates, glassmorphism UI, agent monitoring

API LAYER

FastAPI + WebSockets

RESTful endpoints, async processing, JWT authentication

AI AGENTS

Multi-Agent Orchestration System

8 specialized agents with LLM integration, confidence scoring, consensus building

DATA LAYER

PostgreSQL	pgvector	Apache AGE	Redis
Relational Data	Vector Embeddings	Graph Database	Caching

The 8 Specialized AI Agents

Each Agent Combines LLM Intelligence with Vector Search

Investigation

Primary fraud detection with pattern learning and historical analysis

Compliance

AML, sanctions, and PEP screening with fuzzy matching

Validation

ISO 20022 format validation with anomaly detection

Routing

Intelligent workload distribution and priority assignment

Learning

Continuous improvement from resolution outcomes and feedback

Negotiation

Consensus building when agents have conflicting recommendations

Pattern Detection

Real-time emerging threat identification using vector similarity

Orchestrator

AI-driven coordination strategy and execution sequencing

- **Confidence Scoring** — Every decision includes 0.0-1.0 confidence
- **Human Escalation** — Auto-triggers when confidence < 0.6
- **Full Audit Trail** — Complete logging of all decisions

How It Works

End-to-End Claims Processing

1 Intake → 2 Embed → 3 Analyze → 4 Consensus → 5 Resolve

Parallel Agents

All 8 agents analyze: fraud, compliance, validation

Human Escalation

Auto-triggers when confidence under 0.6

Continuous Learning

Outcomes improve pattern recognition

Intelligent Pattern Detection

How Vector Search Works

1. Embed Every Claim

Convert claim data into high-dimensional vectors

2. Similarity Search

Find similar claims using cosine similarity in pgvector

3. Pattern Recognition

Cluster analysis identifies emerging fraud patterns

4. Continuous Learning

Outcomes update embeddings automatically

Key Capabilities

1536	10ms	10M+
Dimensions	Search Time	Vectors

Fraud Detection

Find similar fraudulent claims, detect mule accounts, identify velocity anomalies

Graph Analysis

Map transaction networks, identify fraud rings, track money flows

Stack: PostgreSQL pgvector + Apache AGE for combined analysis

SWIFT E and I Compliance

Full ISO 20022 Message Support

Supported Message Types

camt.026
Unable to Apply

camt.027
Claim Non-Receipt

camt.028
Additional Info Request

camt.029
Resolution of Investigation

camt.055
Cancellation Request

camt.056
Cancellation Response

camt.036
Debit Authorization Request

camt.037
Debit Authorization Response

Compliance Features

AML Screening

Real-time screening against OFAC, UN, EU, and custom sanctions lists with fuzzy name matching

PEP Detection

Politically Exposed Persons identification with relationship mapping and risk scoring

Audit Trail

Complete decision logging with timestamps, agent reasoning, and confidence scores for regulatory review

ISO Reason Codes

Full support for standard reason codes: AC01, AM04, BE04, CUST, DUPL, FRAD, and 50+ more

Enterprise Dashboard

Claims Management

View, filter, manage claims with real-time WebSocket updates

Agent Monitoring

Live agent status, processing activity, performance metrics

Investigation Details

Agent reasoning, confidence scores, transaction analysis

Human Interventions

Queue management for escalated claims

Analytics

Trends, compliance reports, agent accuracy dashboards

Technology Stack

Production-Ready Enterprise Architecture

Backend

- Python 3.10+
- FastAPI
- Pydantic v2
- WebSockets

Database

- PostgreSQL 14+
- pgvector
- Apache AGE
- Redis

AI / ML

- Chutes LLM Gateway
- OpenAI GPT-4
- Claude / Gemini
- Vector Embeddings

Frontend

- Vanilla JS
- Glassmorphism CSS
- WebSocket Client
- SVG Icons

Async Architecture
Full async/await pattern for non-blocking I/O and maximum throughput

Multi-Model Fallback
Chutes gateway enables automatic fallback between LLM providers

State Management
Centralized Store pattern with path-based access and subscriptions

Error Recovery
Comprehensive error handling with automatic retry strategies

Implementation Roadmap

6-8 Week Phased Approach

1 Foundation
Week 1-2

- Critical bug fixes
- Database schema alignment
- Enum consistency
- WebSocket stability

2 SWIFT Messages
Week 3-4

- camt.055/056 support
- camt.036/037 flows
- Cancellation handling
- Debit authorization

3 API Endpoints
Week 5-6

- Agent status APIs
- Investigation timeline
- Compliance analytics
- Trend reporting

4 Production
Week 7-8

- Performance tuning
- Security hardening
- Load testing
- Documentation

55%
Current Compliance

100%
Target After Phase 4

Key Deliverables
Full message support, complete API coverage

Key Benefits

85%
Faster Processing
From days to seconds

70%
Cost Reduction
Less manual work

95%+
Fraud Detection
AI pattern recognition

5%
False Positives
Precise decisions

Additional: Complete audit trails, regulatory compliance, real-time monitoring, continuous learning, scalable architecture

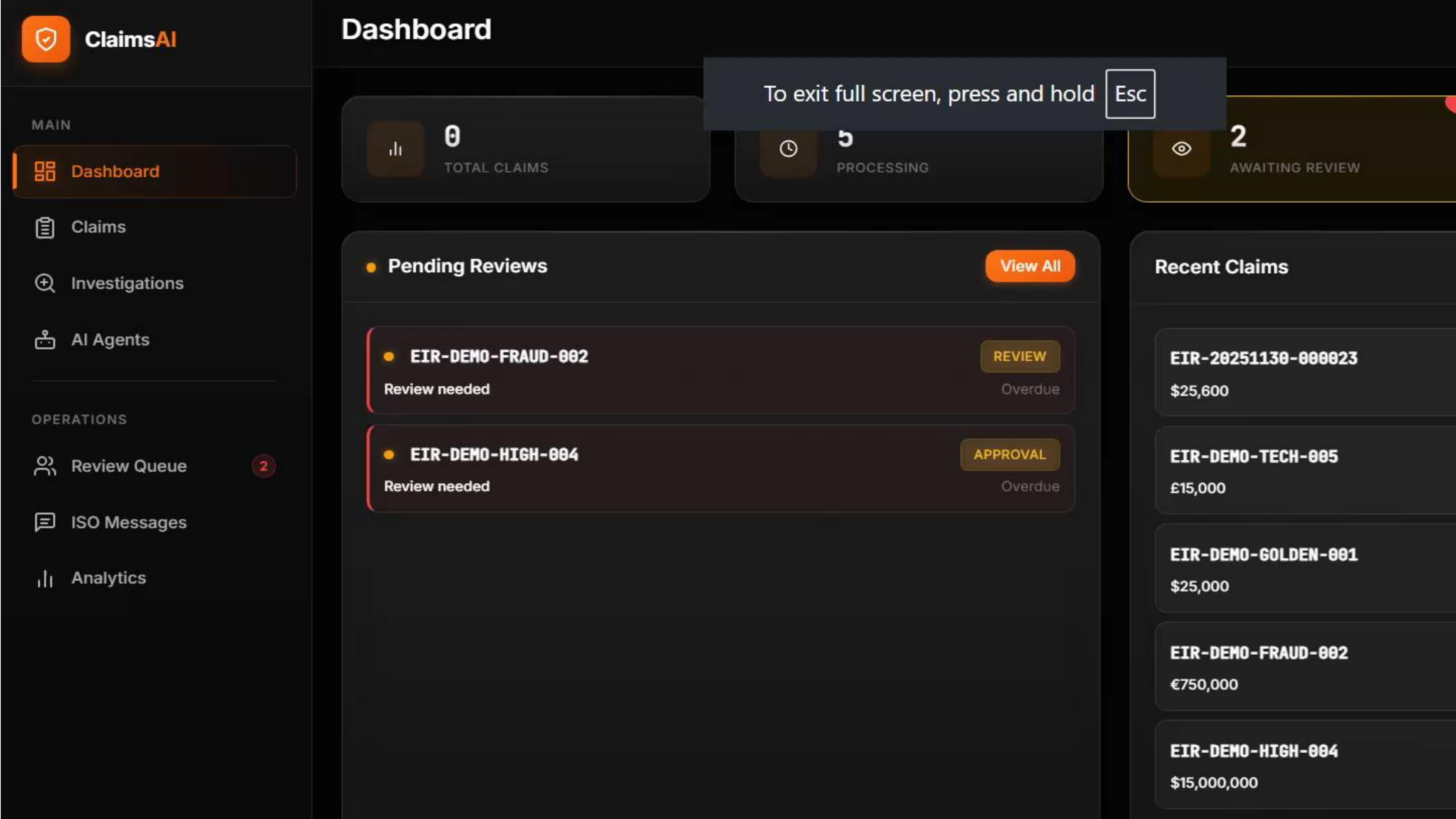
Why This System Is Different

Multi-Agent Intelligence
8 specialized agents collaborate and reach consensus

Vector + Graph Analysis
Semantic similarity combined with network analysis

Human-in-the-Loop
Auto escalation for complex or high-value cases

Self-Improving System
Every resolution improves future accuracy


ClaimsAI
Dashboard
To exit full screen, press and hold Esc
MAIN
Dashboard
Claims
Investigations
AI Agents
OPERATIONS
Review Queue
2
ISO Messages
Analytics
0
TOTAL CLAIMS
5
PROCESSING
2
AWAITING REVIEW
Pending Reviews
View All
EIR-DEMO-FRAUD-002
REVIEW
Review needed
Overdue
EIR-DEMO-HIGH-004
APPROVAL
Review needed
Overdue
Recent Claims
EIR-20251130-000023
$25,600
EIR-DEMO-TECH-005
£15,000
EIR-DEMO-GOLDEN-001
$25,000
EIR-DEMO-FRAUD-002
€750,000
EIR-DEMO-HIGH-004
$15,000,000

AI-NATIVE CLAIMS INVESTIGATION SYSTEM

Thank You

Intelligent fraud detection and compliance screening for the modern financial institution

8	30s	95%	100%
AI Agents	Processing	Detection	Audit Trail